MIL-G-81830A(AS)
28 November 1983
SUPERSEDING
MIL-G-81830(AS)
21 June 1971

MILITARY SPECIFICATION

GROUT MATERIAL, RAPID SETTING

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of rapid setting grout material.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MS 27684	-	Drum, Metal, Shipping and Storage, 3 to 12 Gallons

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR ENGINEERING CENTER

417930	-	Anchor Foot - Lighweight Anchor.
616779	-	Package Assy-F33-Grout Drums.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

FSC 5610

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(Copies of Naval Air Engineering Center drawings may be obtained from the Commanding Officer, Naval Air Engineering Center, Engineering Department (SI), Code 91, Lakehurst, NJ 08733.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the acquisition activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The grout material furnished under this specification shall be products which are qualified for listings on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.3).

3.2 Material. The selection of materials and processes shall be optional with the manufacturer, but shall be restricted by the requirements specified herein. Additionally, the grout material shall not contain any substance which will corrode steel.

3.3 Mixing properties. The grout material shall be sufficiently pre-mixed so that no additional mixing is required other than the liquid required to obtain a pourable state. No more than 2 minutes shall be required for mixing grout mixture and liquid and no special equipment shall be needed, when tested as specified in 4.6.1.

3.4 Gel and set time. When tested as specified in 4.6.3, the gel and set times (in minutes) shall not exceed the following:

Condition	Gel (time)	Set (time)
Normal	15	35
Sea water Mix	10	20
Cold Mix	25	100
Hot Mix	5	60

3.5 Drip test. The maximum acceptable depth penetration shall be 0.090 inch after 24 hours and 0.275 inch after 96 hours when tested as specified in 4.6.4.

3.6 Physical properties. The grout material shall conform to the physical properties listed in Table I.

3.7 Field test. The grout material shall be capable of withstanding a pull test of 20,000 pounds in a properly installed earth anchor, after a setting time of no more than 1 hour. The installed grout shall show no depreciation of physical properties for a period of 1 year after installation.

3.8 Dimensional stability. After mixing, the grout shall not exhibit any shrinkage and expansion shall be limited to 1 percent maximum, when hard.

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3.9 Storage stability. The storage life of the grout material shall be a minimum of 5 years and still retain the characteristics originally required. The grout material shall be no more than 6 months old at time of delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certification of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection. The qualification inspection shall consist of all the tests in Table II.

4.3.1 Sampling instructions. Qualification test samples shall consist of a 100 pound sample in a steel drum. Samples shall be forwarded to the Commanding Officer, Naval Air Engineering Center, Engineering Department (SI), Code 91, Lakehurst, NJ 08733. Sample shall be plainly and durably marked with the following information:

Sample for Qualification Test
Grout Material
Manufacturer's Formula No. _____ and Date Compounded _____
Submitted by (name) (date) for qualification test in accordance with requirements of specification MIL-G-81830A(AS) under authorization (reference authorizing letter) (see 6.3)

4.3.1.1 Field test samples. Field test samples shall consist of 10 - 100 pound drums of grout material. Samples shall be forwarded in accordance with instructions contained in the authorizing letter granting service test which shall be sent to the manufacturer on satisfactory completion of all laboratory tests.

4.3.2 Manufacturer's data. The manufacturer shall submit a report, in duplicate, to accompany the qualification test sample. This report shall include the results of the manufacturer's tests, reported quantitatively, where applicable, in the units specified for all of the requirements specified herein, except storage stability. Tests not conducted due to lack of special test facilities or materials

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shall so be noted in the report. The formulation shall be clearly identified by the manufacturer's formula number.

4.3.3 Retention of qualification. In order to retain qualification of products approved for listing on the qualified products list (QPL), the manufacturer shall verify by certification to the qualifying activity that his product complies with the requirements of this specification. The time of periodic verification by certification shall be in two year intervals from the date of original qualification. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.4 Quality conformance inspection. The quality conformance inspection of grout material shall consist of all the tests in Table II except storage stability and field test.

4.4.1 Lot. A lot shall consist of all grout material manufactured at approximately the same time from the same batch of materials and submitted for acceptance at the same time.

4.4.2 Quality conformance test samples. Quality conformance test samples shall consist of a 100 pound sample in a steel drum. Each sample shall be clearly identified by the manufacturer's formula number as specified in 4.3.2. The test samples shall be forwarded to the Naval Air Engineering Center as specified in 4.3.1. The manufacturer shall furnish with each lot a certificate to the effect that the material has been processed in the same manner and degree using the same base ingredients as the approved qualification sample.

4.4.3 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at Inspection Level I and at Acceptable Quality Level (AQL) of 2.5 percent defective to verify conformance to all requirements of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.5 Inspection conditions. Unless otherwise specified herein, all tests required by this specification shall be conducted at a temperature of 21° to 35°C (70° to 95°F).

4.6 Test methods.

4.6.1 Mixing test. The grout mixture for test coupons shall be based upon the ratio of four gallons of fresh water/sea water, as specified, to one hundred pounds of clean dry grout. The grout mixture shall be weighed accurately, mixed thoroughly and be of consistent uniformity, with ambient temperatures between 21° to 35°C (70° - 95°F).

4.6.2 Test coupons. The test coupons shall be air-cured or water-cured, as specified, for one hour and all tests started within the second hour. The forming of some test coupons will require the mold to be submerged in fresh water/seawater, as directed. No shrinkage and not more than 1 percent expansion shall be permitted.

4.6.3 Gel and set tests.

4.6.3.1 Normal. Accurately measured amounts of dry grout and fresh water shall be stabilized at an ambient temperature of 21° to 35°C (70° to 95°F). The dry grout

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and fresh water shall be properly mixed in an environment of 21° to 35°C (70° to 95°F), then gel and set times shall be recorded. The gel time and set time shall conform to 3.4.

4.6.3.2 Seawater. The preceding test shall also be performed utilizing seawater in lieu of fresh water. Gel and set times shall conform to 3.4.

4.6.3.3 Cold. A measured amount of loose dry grout shall be stabilized at a temperature of -6.7° ± 1°C (20° ± 2°F) and a measured amount of fresh water shall be stabilized at a temperature of 0° ± 1°C (32° ± 2°F) (ice begins to form). The dry grout and fresh water shall be properly mixed in an environment of -6.7° ± 1°C (20° ± 2°F), then gel and set times are to be recorded. The gel and set times shall conform to 3.4.

4.6.3.4 Hot. Accurately measured amounts of dry grout and fresh water shall be stabilized at an ambient temperature of 71° ± 1°C (160° ± 2°F). The dry grout and fresh water shall be properly mixed and gel and set times are to be recorded. The gel and set times shall conform to 3.4.

4.6.4 Drip tests. Fresh water grout mixture and seawater grout mixture drip test specimens shall be prepared and subjected to air-cure. Three test specimens of each type shall be prepared. Each test specimen is to be placed in a test fixture at an angular attitude of 15 degrees from horizontal. Fresh water, at a rate of 30 drops per minute, shall strike the angled face of the test specimen from a height of 18 inches. After 24 hours of test, and again after 96 hours of test, the test specimens shall be measured for depth of penetration using a depth micrometer. The maximum acceptable depth penetration shall not exceed that specified in 3.5.

4.6.5 Compression tests.

4.6.5.1 Air-cure. Fresh water grout mixture shall be poured into square and tubular molds (see Figures 1 and 2), and subjected to air-cure. Three test coupons of each type shall be prepared. Each test coupon shall be centered in a test press and a compression load applied. The compression load shall conform to the value in Table I.

4.6.5.2 Underwater cure. Fresh water grout mixture shall be poured into square and tubular molds (see Figures 1 and 2) and subjected to underwater cure. Three test coupons of each type shall be prepared. Each test coupon shall be centered in a test press and a compression load applied. Conformance to Table I shall be noted.

4.6.5.3 Underwater pour and cure (fresh water). Fresh water grout mixture shall be poured into tubular molds (see Figure 2), submerged in fresh water, through a pouring tube (see Figure 3) and subjected to underwater cure. Three test coupons shall be prepared. After cure, release from the mold and face off the test coupon in a lathe; then test for compression loading. Conformance to Table I shall be noted.

4.6.5.4 Underwater pour and cure (sea water). Seawater grout mixture shall be poured into tubular molds (see Figure 2), submerged in seawater, through a pouring tube (see Figure 3) and subjected to underwater cure. Three test coupons shall be prepared. After cure, release from the mold and face off the test coupon in a

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lathe; then test for compression loading. The minimum acceptable compression load for each coupon shall conform to Table I.

4.6.6 Shear tests.

4.6.6.1 Air-cure. Fresh water grout mixture shall be poured into tubular molds (see Figure 2) and subjected to air-cure. Three test coupons shall be prepared. Each test coupon shall be inserted into a test fixture (see Figure 4), placed in a test press, and a shear load applied. The minimum acceptable shear load for each test coupon shall conform to Table I.

4.6.6.2 Underwater pour and cure. Fresh water grout mixture shall be poured into tubular molds (see Figure 2), submerged in fresh water, through a pouring tube (see Figure 3) and subjected to underwater-cure. Three test coupons shall be prepared. Each test coupon shall be inserted into a test fixture (see Figure 4), placed in a test press, and a shear load applied. The minimum acceptable shear load for each test coupon shall conform to Table I.

4.6.6.3 Underwater pour and cure. Sea water grout mixture shall be poured into tubular molds (see Figure 2), submerged in sea water, through a pouring tube (see Figure 3) and subjected to underwater cure. Three test coupons shall be prepared. Each test coupon is to be inserted into a test fixture (see Figure 4), placed in a test press, and a shear load applied. The minimum acceptable shear load for each test coupon shall conform to Table I.

4.6.7 Tension tests. Fresh water grout mixture shall be poured into tension test mold (see Figure 5) and subjected to air-cure. Three test coupons shall be prepared. Each test coupon is to be placed in a test press and subjected to a test load. The minimum acceptable tension test load for each test coupon shall conform to Table I.

4.6.8 Adherence tests. Fresh water grout mixture shall be poured into the mold (see Figure 6) and subjected to air-cure. Three test coupons shall be prepared. Each test coupon shall be pull-tested to determine withdrawal loads. The minimum acceptable withdrawal load for each test coupon shall conform to Table I.

4.6.9 Bond tests.

4.6.9.1 Air-cure. Fresh water grout mixture shall be poured into the test fixture (see Figure 7) and subjected to air-cure. Three test coupons shall be prepared. Each test coupon shall be pull-tested to determine withdrawal loads. The withdrawal load for each test coupon shall conform to Table I.

4.6.9.2 Underwater pour and cure (fresh water). Fresh water grout mixture shall be poured into the test fixture (see Figure 7), submerged in fresh water, through a pouring tube (see Figure 3) and subjected to underwater-cure. Three test coupons shall be prepared. Each test coupon shall be pull-tested to determine withdrawal loads. The minimum acceptable withdrawal load for each test coupon shall be in accordance with Table I.

4.6.9.3 Underwater pour and cure (seawater). Seawater grout mixture shall be poured into the test fixture (see Figure 7), submerged in seawater, through a pouring tube (see Figure 3) and subjected to under water cure. Three test coupons shall be prepared. Each test coupon shall be pull-tested to determine withdrawal

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loads. The withdrawal load for each test coupon shall be in accordance with Table I.

4.6.10 Field test. The Government shall provide the earth anchors, installation services and material, and perform the required tests. The field test (3.7) shall be performed when grout materials have met all the requirements of Section 3 except storage stability.

4.6.11 Dimensional stability. Dimensional stability (3.8) shall be certified by the manufacturer (see 4.1.1).

4.6.12 Storage stability. The storage stability (3.9) shall be certified by the manufacturer (see 4.1.1).

5. PACKAGING

5.1 Application. The packaging, packing, and marking requirements specified in Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging. The grout material shall be packaged in quantities of 100 pounds in a heat sealed, heavy duty plastic bag, in a metal drum conforming to MS27684-8, 12 gallon capacity and cover shall be in accordance with MS27684-24, Style 2.

5.3 Packing.

5.3.1 Individual drums. Individual drums (see 5.2), require no additional packing.

5.3.2 Palletizing. Individual drums may be palletized, when specified, 10 per pallet in accordance with Drawing 616779.

5.4 Marking. Each container shall be marked in accordance with MIL-STD-129 and marking shall include the following:

a. Title and number of this specification

b. Net weight of contents

c. Date of manufacture

d. Name of manufacturer

e. Contract number

6. NOTES

6.1 Intended use. The grout material covered by this specification is intended for use with the Lightweight Earth Anchor - LEA-20 (see Drawing 417930).

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title and number of this specification

b. Quantity desired in pounds

c. Packaging, packing and marking data with requirements in detail, if other than as specified in Section 5.

d. Whether individual drums or palletized drums are required (see 5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL-81830), whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Engineering Department (SI), Code 91, Naval Air Engineering Center, Lakehurst, NJ 08733. Information pertaining to qualification of products and letter of authorization may be obtained from that activity.

6.3.1 Product information. The grout material furnished under contract shall be identical in every respect to the qualification samples which have been inspected and approved. In the event that the grout material furnished under contract is found to deviate from the properties of the approved product or that the product fails to perform satisfactorily, approval of such product will be subjected to immediate withdrawal from the Qualified Products List.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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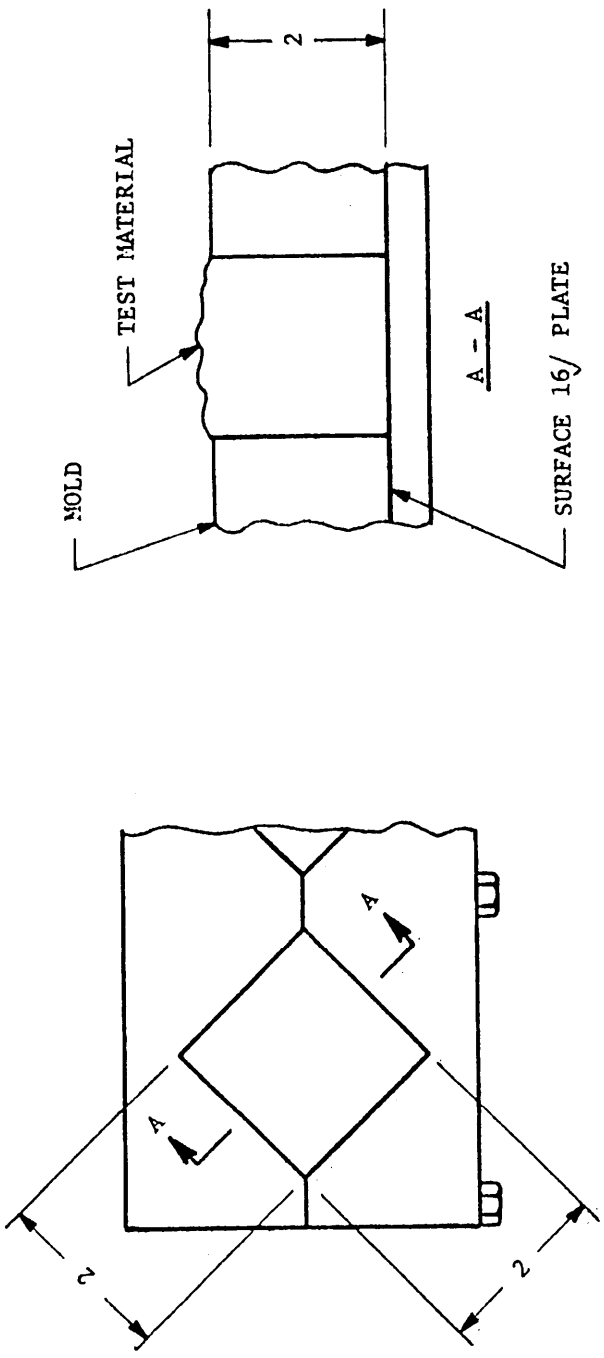
TABLE I. Physical properties.

Characteristic	Requirement				Test Paragraph
	Air cure	Fresh water cure	Fresh water pour & cure	Seawater pour & cure	
Compression test, psi-min.	2,500	2250	1250	1000	4.6.5
Shear test, psi-min.	700		400	50	4.6.6
Tension test, psi-min.	75				4.6.7
Adherence test, lbs.-min.	2,500				4.6.8
Bond test, lbs.-min.	35,000		20,000	9000	4.6.9

TABLE II. Qualification inspection.

Characteristic	Paragraph	
	Requirement	Test
Mixing properties	3.3	4.6.1
Gel and set time	3.4	4.6.3
Drip test	3.5	4.6.4
Compression test	table I	4.6.5
Shear test	table I	4.6.6
Tension test	table I	4.6.7
Adherence test	table I	4.6.8
Bond test	table I	4.6.9
Field test	3.7	4.6.10
Dimensional stability	3.8	4.6.11
Storage stability	3.9	4.6.12

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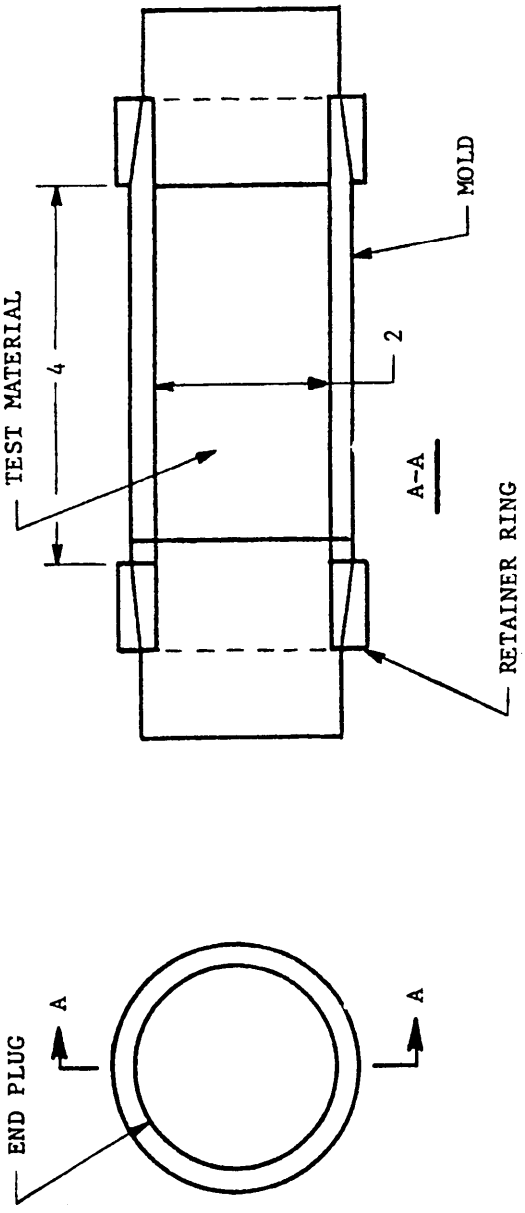


NOTES:

1. COAT INSIDE OF MOLD WITH HARD WAX BEFORE POURING TEST SAMPLE.
2. DIMENSION IN INCHES, UNLESS OTHERWISE SPECIFIED.

SQUARE MOLD
FIGURE 1.

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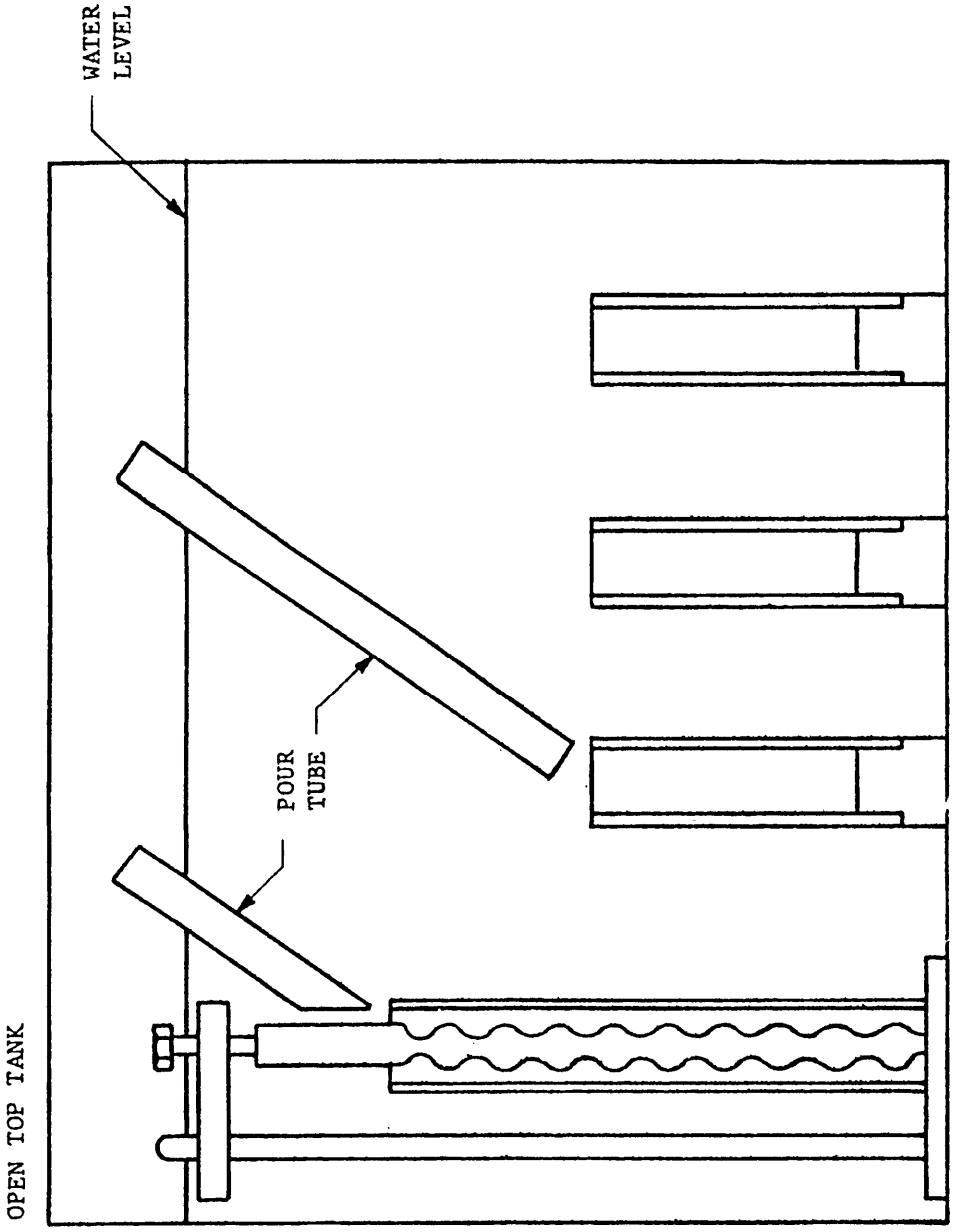


NOTES:

1. MOLD IS SPLIT LENGTHWISE TO FACILITATE REMOVAL OF TEST SAMPLE.
2. COAT INSIDE OF MOLD WITH HARD WAX BEFORE POURING TEST SAMPLE.
3. DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.

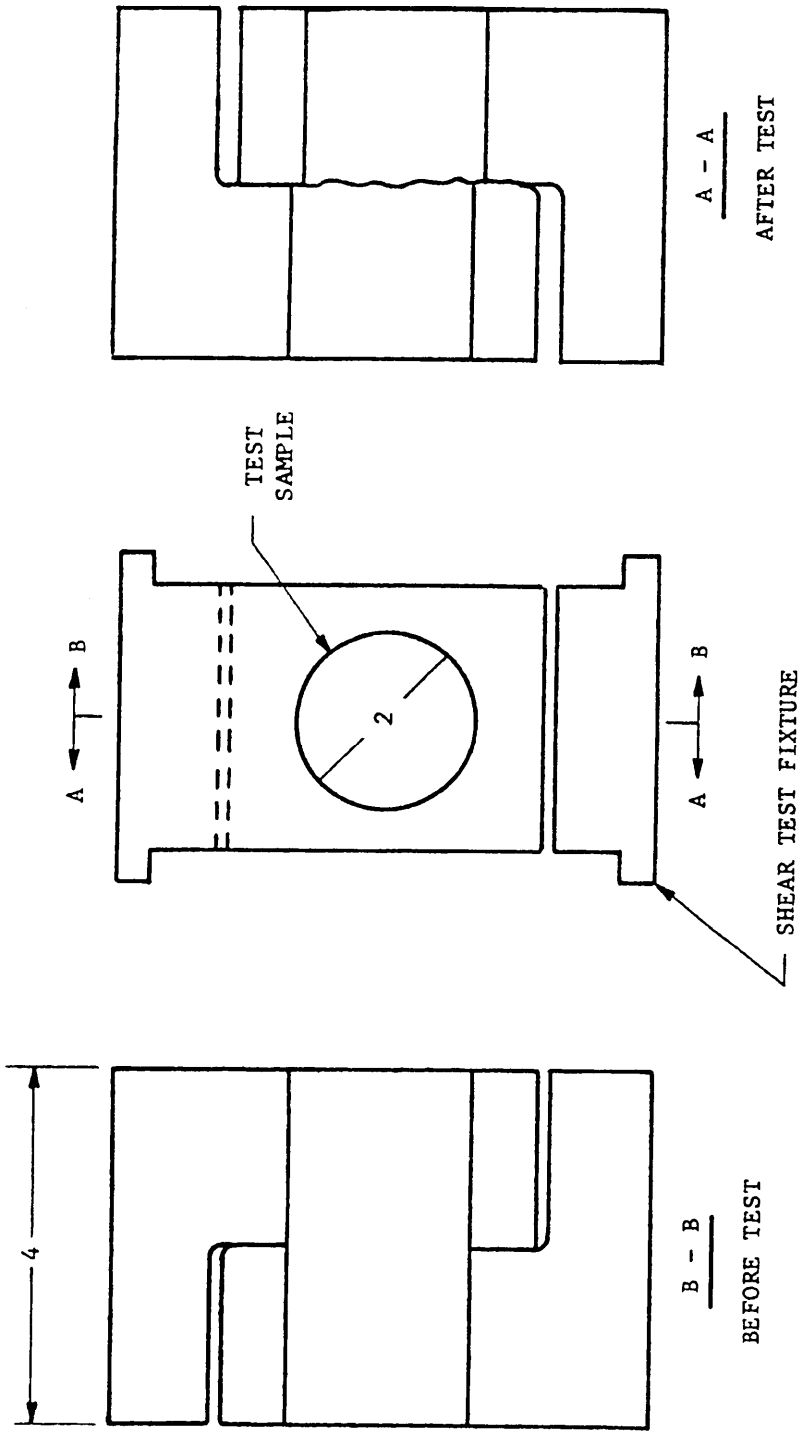
TUBULAR MOLD
FIGURE 2.

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UNDERWATER POUR AND CURE FACILITY
FIGURE 3.

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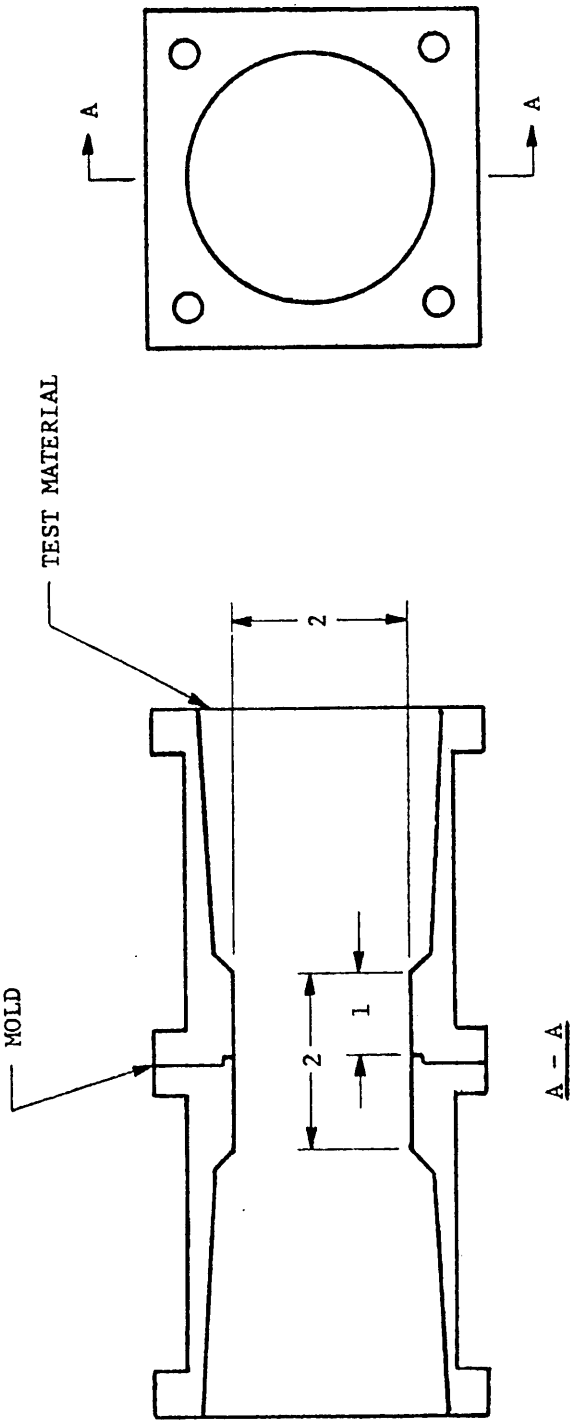


SHEAR TEST FIXTURE
FIGURE 4.

NOTE:

1. DIMENSION IN INCHES, UNLESS OTHERWISE SPECIFIED.

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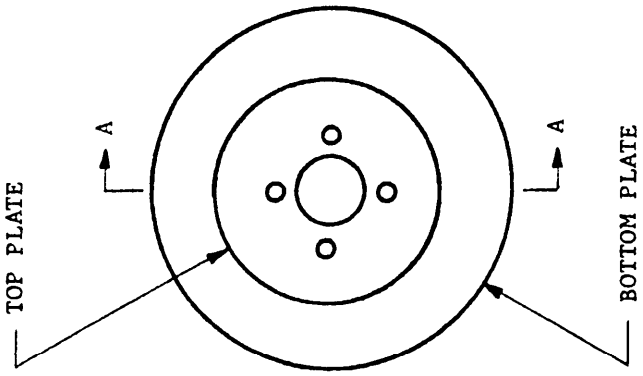


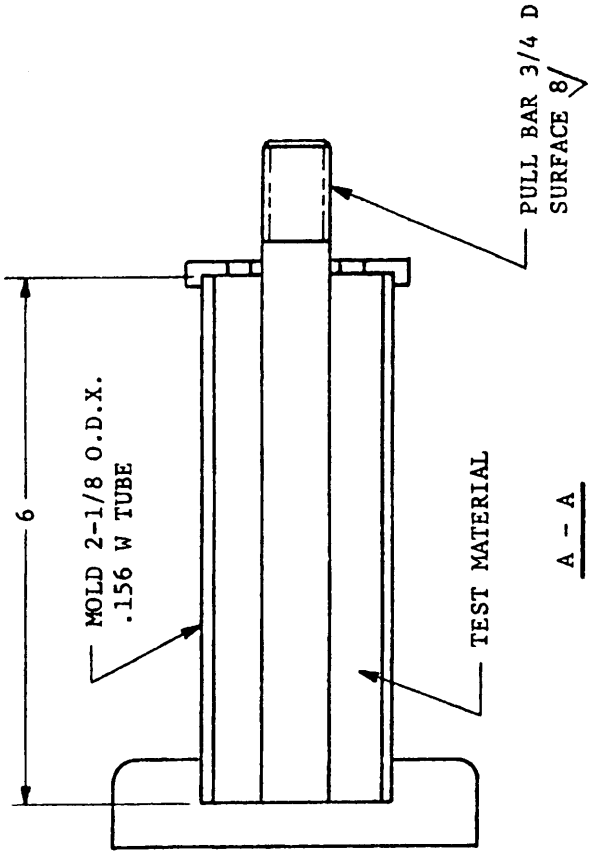
TENSION TEST MOLD
FIGURE 5.

NOTES:

1. SURFACE 16/ INSIDE OF MOLD.
2. DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.

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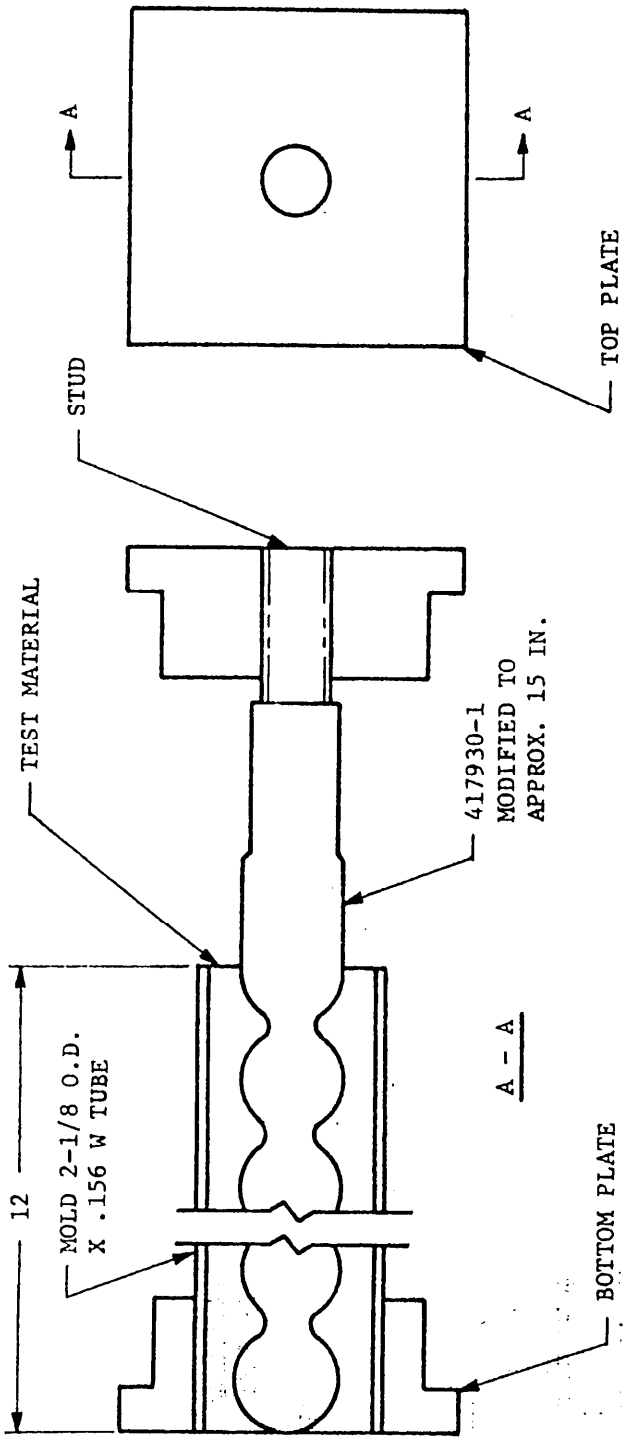
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NOTE:

1. DIMENSION IN INCHES, UNLESS OTHERWISE SPECIFIED.

ADHERENCE TEST MOLD
FIGURE 6.

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BOND TEST FIXTURE
FIGURE 7.

NOTE:

1. DIMENSION IN INCHES, UNLESS OTHERWISE SPECIFIED.

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c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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